


C-CLARKE
INTERNATIONAL INSTITUTE

# BECOME AN

# ADVANCED
# AI AND ML DEVELOPER

**8 Months Program**

Get Certified From
The Official Python Education Center

PYTHON INSTITUTE
OPEN EDUCATION & DEVELOPMENT GROUP

**Contact Details :**

www.c-clarke.com  info@c-clarke.com  +94 70 1 000000

## TABLE OF CONTENTS


1. Welcome Message
2. About us
3. Why This Program Changes Everything
4. Panel Of Lecturers And Academic Consultants
5-6. Curriculum Overview
7. Course Fees and Payment Options
8. Contact Information




## WELCOME MESSAGE



Dear Students,

**Welcome to C-Clarke International Institute of Digital Sciences!**

We are thrilled to have you join us on this exciting journey of learning, innovation, and professional growth with us. At C-Clarke, we are dedicated to providing you with an exceptional education that combines advanced technologies, practical skills, and industry insights to prepare you for a successful career in software landscape.

Our institute is more than just a place to study—it's a community where connections are made, ideas are shared, and futures are shaped. Guided by a team of esteemed lecturers and supported by strong industry partnerships, we are committed to helping you achieve your goals and become leaders who make meaningful contributions both locally and globally.

We encourage you to take full advantage of the resources, opportunities, and support available at C-Clarke. Together, let's build a brighter future for you and our society.

Wishing you great success

*Director, Academic Affairs*

## ABOUT US

C-Clarke International Institute Of Digital Sciences is a premier educational institution dedicated to shaping the future Of software technology professionals. With a commitment to delivering innovative, industry-relevant education, we enable students to excel in the fast-evolving world of software and digital sciences.

Our programs are precisely designed to provide a strong foundation in advanced technologies, practical Skills, and ethical practices, ensuring our students are equipped for success both locally and globally. Backed by a distinguished panel Of lecturers and consultants with extensive industry experience, we offer unparalleled opportunities for learning and networking.

At C- Clarke, we prioritize not just education but also the holistic development of our students. By fostering innovation, growth, and lifelong learning, we aim to create a community of future leaders who drive progress, achieve economic independence, and contribute meaningfully to the world.

C-Clarke is not just a place to study; it's a place to connect, grow and transform your aspirations into reality.



## VISION AND MISSION

**Vision**

To Be acknowledged as the Leader in Education for Digital Sciences, Whilst Inspiring Young Leaders to Achieve Economic Strength and Independence.

**Mission**

To Provide Innovative Education That Builds Essential Skills for Professional Success Locally and Globally.

## CORE VALUES

**Innovation**

Continuously adhering to the latest global technologies, ensuring our programs stay ahead of industry trends to prepare students for the future.

**Integrity**

Committed to transparency, honesty, and ethical practices, fostering trust and respect in all our educational endeavors.

**Excellence**

Striving for the highest standards in education and guiding students to reach their full potential through world-class learning experiences.

**Empathy**

Building a supportive and compassionate community that understands the personal and professional aspirations of each student, guiding them with care and dedication toward their success

## Why This Program Changes Everything

### Train Your Brain for the AI Era

This isn't about memorizing frameworks—it's about fundamentally rewiring how you think about problem-solving in an AI-first world. You'll develop the mental models that separate AI practitioners from AI masters.

### Build Real Products, Not Toy Projects

Forget Hello World tutorials. You'll ship production grade AI applications that can handle real users, real data, and real business challenges. Your portfolio will speak louder than any certificate.





### Dual Learning Modes

Choose in-person immersion for maximum collaboration and networking, or online flexibility to learn from anywhere. Both options deliver the same rigorous curriculum and industry-recognized certification.

### Six Months to Career Transformation

We've compressed what traditional programs drag out over years into an intensive, focused sprint. Every week brings new capabilities. Every month brings career-defining projects. Six months brings total transformation.

C-CLARKE INTERNATIONAL INSTITUTE

# PANEL OF LECTURERS AND ACADEMIC CONSULTANTS



## SUSHAIN DILISHAN

BSc (Hons) in
Software Engineering
Cardiff Metropolitan University

Tech Lead at
Sysco Labs



## PRASHAN SILVA

BSC (Hons) in
Computer Science
University Of Kelaniya

Tech Lead at
CodeGen



## DILSHAN ARIYARATHNA

BSc (SP) in
Computer Science
University of Kelaniya

Associate Technical
Lead Engineer at
iCan Lanka



## CHAMINDIKA KODITHUWAKKU

Master of Information
Technology
University of Colombo
BSc. (Hons) in
IT & Management
University of Moratuwa

Senior Software Engineer
CMS

# CURRICULUM OVERVIEW

## Module 1: Professional Python Development (Python)

1.1 Introduction to Python & Development Environment
1.2 Python Syntax, Variables & Data Types
1.3 Operators & Expressions
1.4 Control Flow Statements (if, elif, else)
1.5 Looping Structures (for, while)
1.6 Functions & Lambda Expressions
1.7 Core Data Structures (Lists, Tuples, Sets, Dictionaries)
1.8 Advanced Data Structures & Comprehensions
1.9 Object-Oriented Programming — Fundamentals
1.10 Object-Oriented Programming — Advanced Concepts
1.11 Modules & Packages
1.12 Exception Handling & Custom Exceptions
1.13 File Handling & File Operations
1.14 File Handling & File Operations - Day 2
1.15 Regular Expressions (Regex)
1.16 Debugging Techniques & Code Testing

## Module 2: Introduction to Web Development (Web)

2.1 Introduction to Web Development & Web Architecture
2.2 HTML5 Fundamentals (Structure & Semantics)
2.3 HTML5 Forms, Media & Best Practices
2.4 CSSS Fundamentals (Selectors, Colors, Units)
2.5 CSS Box Model & Layout Techniques
2.6 CSS Flexbox
2.7 CSS Grid & Responsive Design Principles
2.8 Advanced CSS3 (Animations, Transitions) JavaScript ES6 Fundamentals
2.10 JavaScript Control Flow & Functions
2.11 DOM Manipulation & Events
2.12 JavaScript ES6 Advanced Features
2.13 Ajax Concepts & Asynchronous Programming
2.14 Fetch API & JSON Handling
2.15 Introduction to JavaScript Libraries
2.16 Bootstrap Framework
2.17 Tailwind CSS Framework

## Module 3: FastAPI Backend Development

3.1 Introduction to Web APIs & REST Concepts
3.2 FastAPI Framework Introduction
3.3 Building REST APIs with FastAPI
3.4 Request Handling, Path Parameters & Query Parameters
3.5 Data Validation using Pydantic Models
3.6 CRUD API Development with FastAPI
3.7 FastAPI Database Integration (SQLAlchemy)
3.8 API Documentation (Swagger / OpenAPI)
3.9 Authentication & Security Basics (JWT)
3.10 Async Programming in FastAPI

## Module 4: Advanced AI Development (AI)

4.1 Introduction to Artificial Intelligence & AI Ecosystem
4.2 Mathematics for AI — Linear Algebra
4.3 Mathematics for AI - Probability & Statistics
4.4 Data Handling & Data Collection
4.5 Data Cleaning, Preprocessing & Feature Engineering
4.6 Introduction to AI APIs & Frameworks
4.7 Machine Learning Fundamentals
4.8 Regression Algorithms
4.9 Classification Algorithms
4.10 Model Evaluation & Performance Metrics
4.11 Introduction to Deep Learning
4.12 Neural Networks & Backpropagation
4.13 Deep Learningwith AI APIs
4.14 Convolutional Neural Networks (CNNs)
4.15 Natural Language Processing (NLP) — Fundamentals
4.16 NLP Applications
4.17 Real-World AI Applications & Case Studies
4.18 Ethics in AI & Responsible AI
4.19 AI Deployment Concepts
4.20 AI DevOps & MLOps

# CURRICULUM OVERVIEW

**Module 5: Prompt Engineering (Prompt)**

5.1 Introduction to Large Language Models (LLMs)
5.2 Capabilities, Limitations & use Cases of LLMs
5.3 Introduction to Prompt Engineering
5.4 Prompt Structure & Components
5.5 Zero-Shot Prompting Techniques
5.6 Few-Shot Prompting Techniques
5.7 Chain-of-Thought & Advanced Prompting Strategies
5.8 Improving Reliability, Consistency & Prompt Optimization

**Module 6: LLM Engineering (LLM)**

6.1 Introduction to LLM Engineering & Ecosystem
6.2 Overview of Open-Source vs Proprietary LLMs
6.3 GPT, Claude, LLaMA & Mistral Model Families
6.4 LLM Architecture Basics (Transformer Overview)
6.5 LLM Inference Concepts
6.6 LLM Inference with APIs
6.7 Local LLM Inference using Ollama
6.8 Prompt Engineering vs Fine-Tuning
6.9 Fine-Tuning Concepts & Use Cases
6.10 Parameter-Efficient Fine-Tuning (LoRA, PEET)
6.11 Introduction to Retrieval-Augmented Generation (RAG)
6.12 RAG Architecture & Workflow
6.13 Vector Databases & Embeddings
6.14 Document Ingestion & Chunking Strategies
6.15 LangChain Fundamentals
6.16 LangChain Chains, Tools & Memory
6.17 LlamaIndex Fundamentals & Data Connectors
6.18 Custom Tool Integration & Function Calling
6.19 Building AI Agents with Tools & Memory
6.20 Autonomous Agent Architectures & Workflows
6.21 Guardrails & Safety in LLM Applications

**Module 7: Database Management Systems (Integrated with JAVA)**

7 .1 Database Models & Architecture
7.2 Relational Database Concepts
7 .3 SQL Fundamentals
7.4 Advanced SQL & Performance Optimization
7 .5 NoSQL Databases
7.6 Database Design, Security & Administration

**Module 8: Comprehensive Angular Training (Angular)**

8.1 Introduction to Angular & Architecture
8.2 Components, Templates & Data Binding
8.3 Directives & Pipes
8.4 Services & Dependency Injection
8.5 Routing & Navigation
8.6 HTTP Client & Observables (RxJS)
8.7 State Management & Advanced RxJS Concepts
8.8 Testing & Debugging in Angular

**Module 9: Version Control (GIT)**

9.1 Introduction to Version Control & Git Basics
9.2 Git Workflow, Branching, Merging & GitHub Collaboration

# COURSE FEES AND PAYMENT OPTIONS

COMMERCIAL BANK

Account Name : C-Clarke International Institute of Digital Sciences
Bank: Commercial Bank Bank Code: 7056
Branch: Moratuwa Branch Code: 59
Account Type: Current Account
Account Number : 1000874493

PAN ASIA BANK

Account Name : C-Clarke International Institute of Digital Sciences
Bank: Pan Asia Bank Bank Code: 7311
Branch: Moratuwa Branch Code: 024
Account Type: Current Account
Account Number : 102411001581

Please Note:

- All completed payments are non-refundable
- We encourage all students to reach out for guidance to ensure a smooth payment process.

Important Instructions

- When making the payment, please include your NIC number on the deposit slip or in the remarks section for online transfers.
- If you require specific payment instructions, contact your Student Counselor before proceeding
- Hotline: 0701000000

## ADVANCED AI & ML PROGRAM

**Course fee :** 120,000 LKR
**Early Bird Offer :** 20,000 off for one time payments.
**One Time Fee - 100,000 LKR**
**Duration : 08 Months**



## REFUND POLICY

At C-Clarke International Institute of Digital Sciences, we maintain a strict **NO-REFUND POLICY** for all completed payments. Once a payment is made, it is considered final and non-refundable.

We strongly encourage all prospective students to review the program details, schedules, and payment terms carefully before proceeding with the payment. For any clarifications, our student counselors are available to assist you prior to confirming your enrollment.

This policy ensures the smooth operation of our programs and the continued delivery of high-quality education to all enrolled students.

## CONTACT INFORMATION


| | | |
|---|---|---|
| **Address** | : | 517 1/1 Galle Road, Rawatawatta, Moratuwa. |
| **Phone** | : | 0701 000 000 (Call/WhatsApp) |
| **Email** | : | info@c-clarke.com |
| **Website** | : | www.c-clarke.com |
| **Linked In** | : | https://www.linkedin.com/company/c-clarke |
| **Facebook** | : | http://www.facebook.com/cclarkecampus |
| **Instagram** | : | http://www.instagram.com/cclarkecampus |